Image Processing Application: Comprehensive UI Features and Operations Report

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***Abstract*—This report details the graphical user interface (GUI) and operational features of a comprehensive computer vision application designed for image processing tasks. The application features a modular and dynamic interface that allows users to apply various algorithms, including noise generation, spatial filtering, edge detection, histogram operations, and color space transformations.**

***Index Terms*—Image Processing, Graphical User Interface, Computer Vision, Edge Detection, Filtering, Histogram, Hybrid Images.**

I. Introduction

The "CV Assignment - Image Processing Tasks" application features a comprehensive user interface designed for applying various image processing algorithms. The standard layout is logically divided into visual display panels and a control panel on the right side of the window.

II. Supported Image Processing Tasks

A. Noise Addition

The UI supports introducing synthetic noise models into the original image to test robustness or simulate sensor artifacts.

Uniform noise. Selecting "Uniform" in the "Noise" dropdown exposes a single *Noise amplitude* parameter, set to 74 here. After pressing *Run Operation*, the noisy image appears in Box 2, which is the active output box.

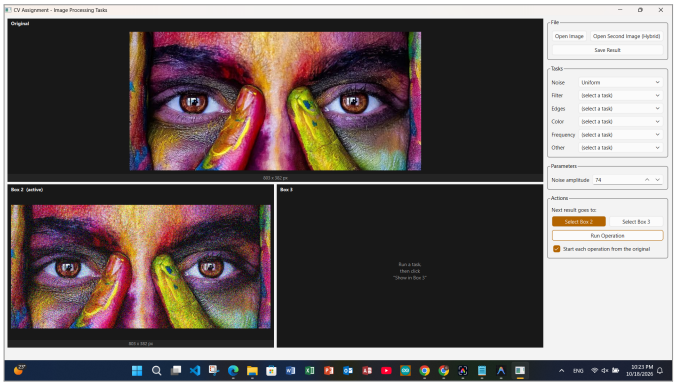

Fig. 1: Uniform Noise applied to the image.

Gaussian noise. Selecting "Gaussian" replaces the amplitude control with a *Gaussian mean* (0.00) and a *Gaussian stddev* (45.00). This time the output is directed to Box 3, so Box 2 still shows the previous uniform result and the two noise models can be compared side by side.

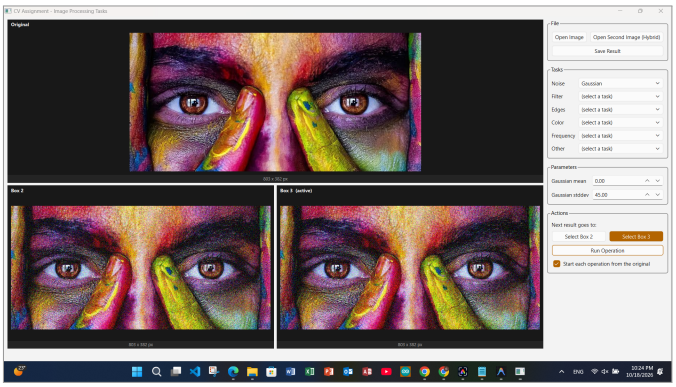

Fig. 2: Gaussian Noise applied with a standard deviation of 45.00.

Salt & pepper noise. Selecting "Salt & Pepper" exposes a single *Noise probability* parameter, set to 0.200 here, which controls the fraction of pixels that are corrupted. The affected pixels are replaced by pure black or pure white values, giving the scattered speckle pattern visible in Box 2. A grayscale test image is used in this example.

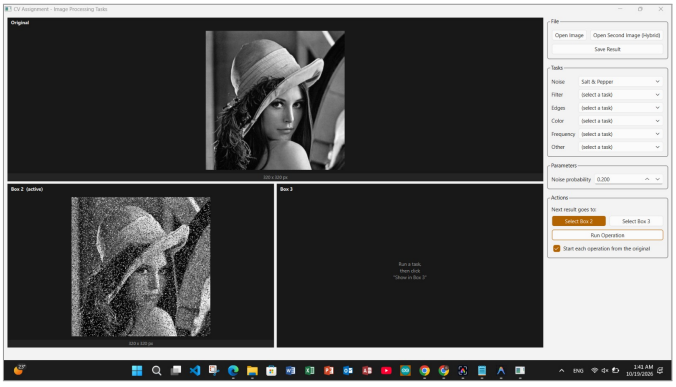

Fig. 3: Salt & Pepper Noise applied with a noise probability of 0.200.

B. Spatial Filtering Techniques

Spatial filtering operations can be selected from the "Filter" dropdown menu for smoothing and blurring.

Average filter. The average filter replaces each pixel by the mean of its neighborhood, and its only parameter is the *Kernel*

size. With a 3×3 kernel the result in Box 2 is visibly smoother than the noisy image still shown in Box 3.

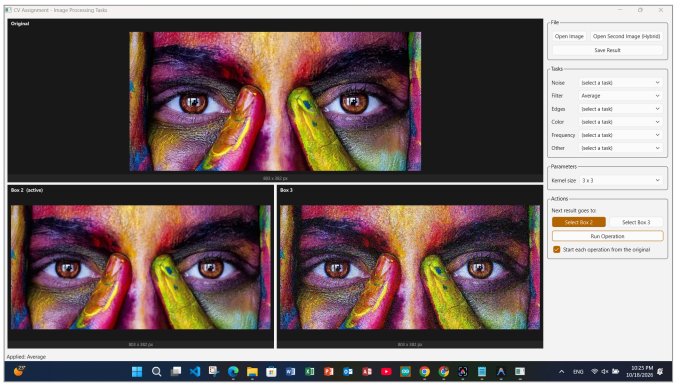

Fig. 4: Average Filter applied with a 3x3 kernel.

Gaussian filter. The Gaussian filter adds a *Gaussian sigma* parameter next to the kernel size, which controls how quickly the weights fall off from the center. Here a 3×3 kernel with sigma 1.0 is applied, and the result is sent to Box 3.

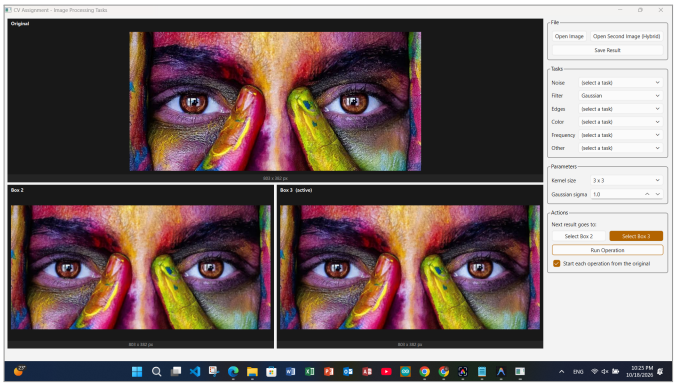

Fig. 5: Gaussian Filter applied with a 3x3 kernel and sigma 1.0.

C. Edge Detection Algorithms

The application supports multiple gradient and edge detection operators under the "Edges" task category.

Prewitt (X direction). The Prewitt operator has a single *Direction* parameter. With "X direction" selected, only the gradient along the X axis is computed, so the output highlights intensity changes in that direction.

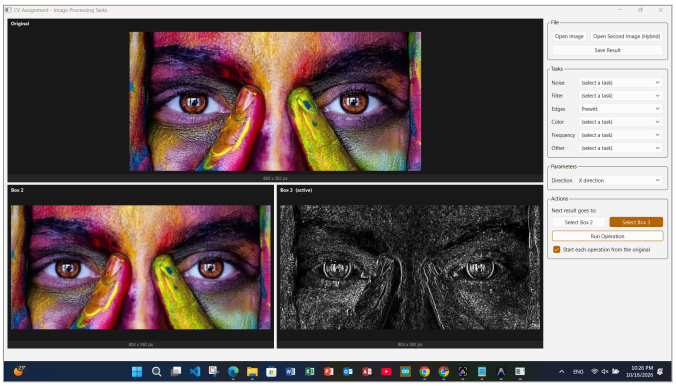

Fig. 6: Prewitt Operator (X direction).

Prewitt (X and Y). Choosing "X,Y directions" combines the horizontal and vertical gradients into one edge map, which shows the outlines of the eyes, fingers, and skin texture more completely than the single-direction result.

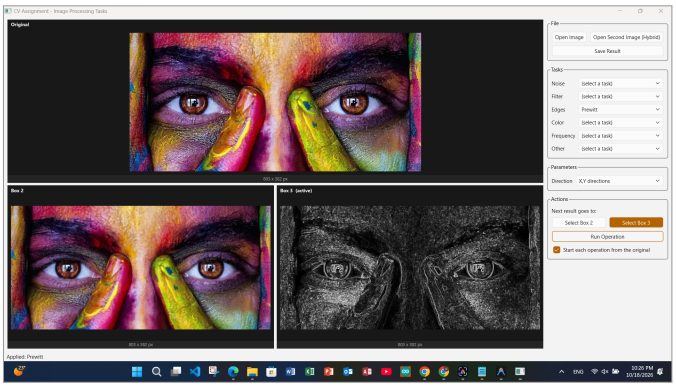

Fig. 7: Prewitt Operator (X and Y directions combined).

Canny (thresholds 50 and 150). The Canny detector exposes a *Canny low threshold* and a *Canny high threshold*. With 50 and 150, many weak edges are kept, which produces a dense edge map in Box 2.

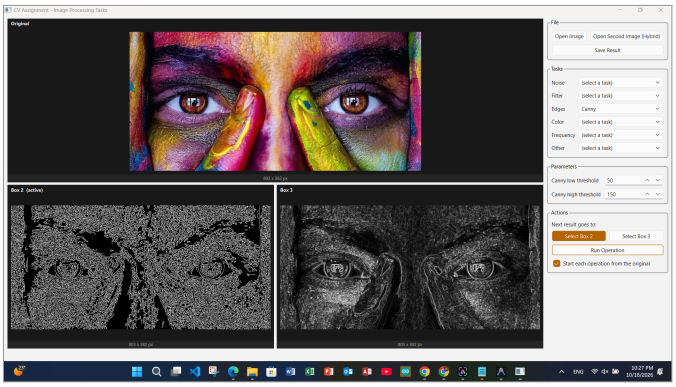

Fig. 8: Canny Edge Detector with low threshold 50 and high threshold 150.

Canny (low threshold 96). Raising only the low threshold to 96 while keeping the high threshold at 150 discards more weak edges, so the edge map in Box 2 becomes noticeably sparser and cleaner.

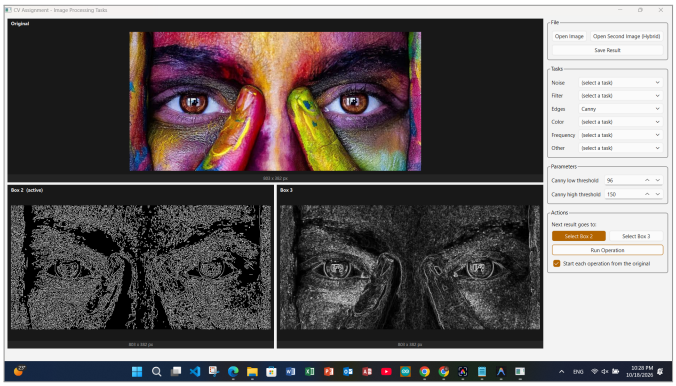

Fig. 9: Canny Edge Detector with adjusted low threshold of 96.

D. Histogram Operations

The application includes dedicated features for analyzing and enhancing the tonal distribution of images.

- **Histogram Plotting**: The interface allows users to compute and visualize the frequency distribution of pixel intensities across the image.
- **Histogram Equalization**: Users can apply equalization to improve the global contrast of the image, which effectively spreads out the most frequent intensity values.

Histogram plotting. Selecting "Histogram" in the "Other" dropdown needs no parameters. The plot appears in Box 2, with gray values from 0 to 255 on the horizontal axis and the pixel count on the vertical axis.

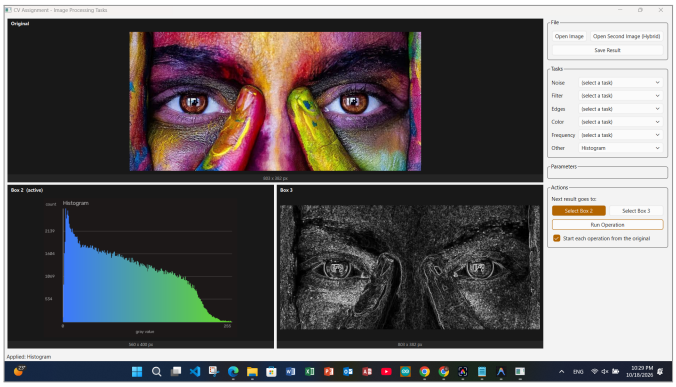

Fig. 10: Histogram of the image gray values.

Histogram equalization. Selecting "Equalization" also needs no parameters. The equalized image in Box 2 spreads the intensity values more evenly, which improves the global contrast compared with the original.

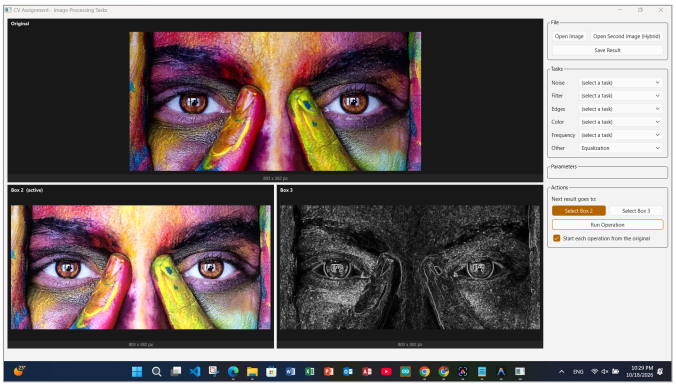

Fig. 11: Histogram Equalization applied to the image.

E. Color Space Transformations

Users can manipulate the color space of the image using the "Color" task dropdown.

Grayscale. Selecting "Transfer to Grayscale" converts the color image into a single-channel gray image. This operation has no parameters, so the *Parameters* group stays empty.

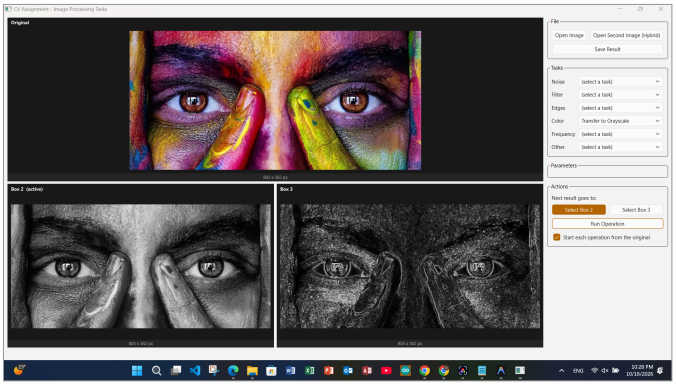

Fig. 12: Transfer to Grayscale operation.

F. Other Operations (Normalization)

The application includes an "Other" category for miscellaneous image processing routines such as normalization.

Normalization. Selecting "Normalization" also needs no parameters. The result is placed in Box 2, and the status bar at the bottom of the window confirms the operation with "Applied: Normalization".

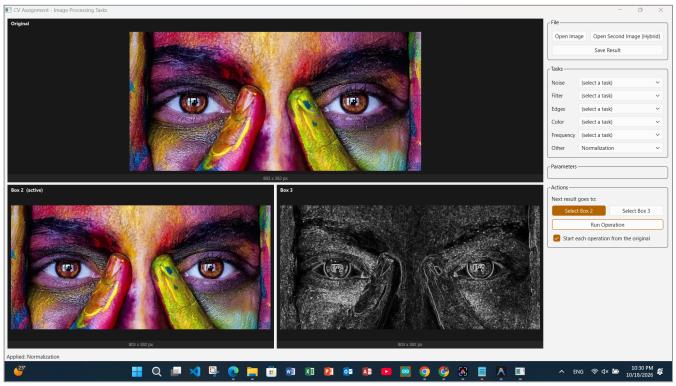

Fig. 13: Image Normalization applied, with the result mapped to Box 2.

III. Hybrid Image Generation Module

Selecting "Hybrid Image" from the "Other" dropdown significantly alters the application's layout, shifting from a single-image pipeline to a dual-image processing framework. Users can independently apply a Low Pass filter to Image 1 and a High Pass filter to Image 2.

Hybrid layout. The window now shows both input images at the top, a filter panel for each image (filter type and a cutoff slider with an *Apply Filter* button), the two filtered images below, and the final *Hybrid Result* at the bottom after pressing *Create Hybrid Image*. In this example, Image 1 is low-pass filtered with cutoff 12 and Image 2 is high-pass filtered with cutoff 47. The result keeps the smooth structure of Image 1 and adds the fine detail of Image 2.

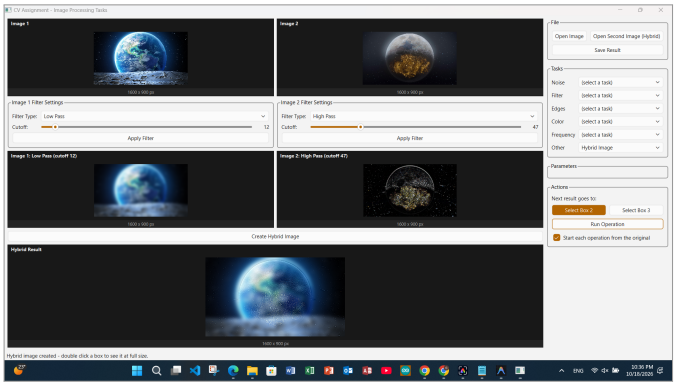

Fig. 14: Hybrid Image generation layout showing the Low Pass and High Pass cutoffs and the final result.

IV. Conclusion

The software delivers an intuitive, modular UI where algorithm parameters dynamically update based on the selected computer vision task. The interface proves highly adaptable, seamlessly transitioning into a specialized layout to handle complex multi-image procedures like Hybrid Image generation.